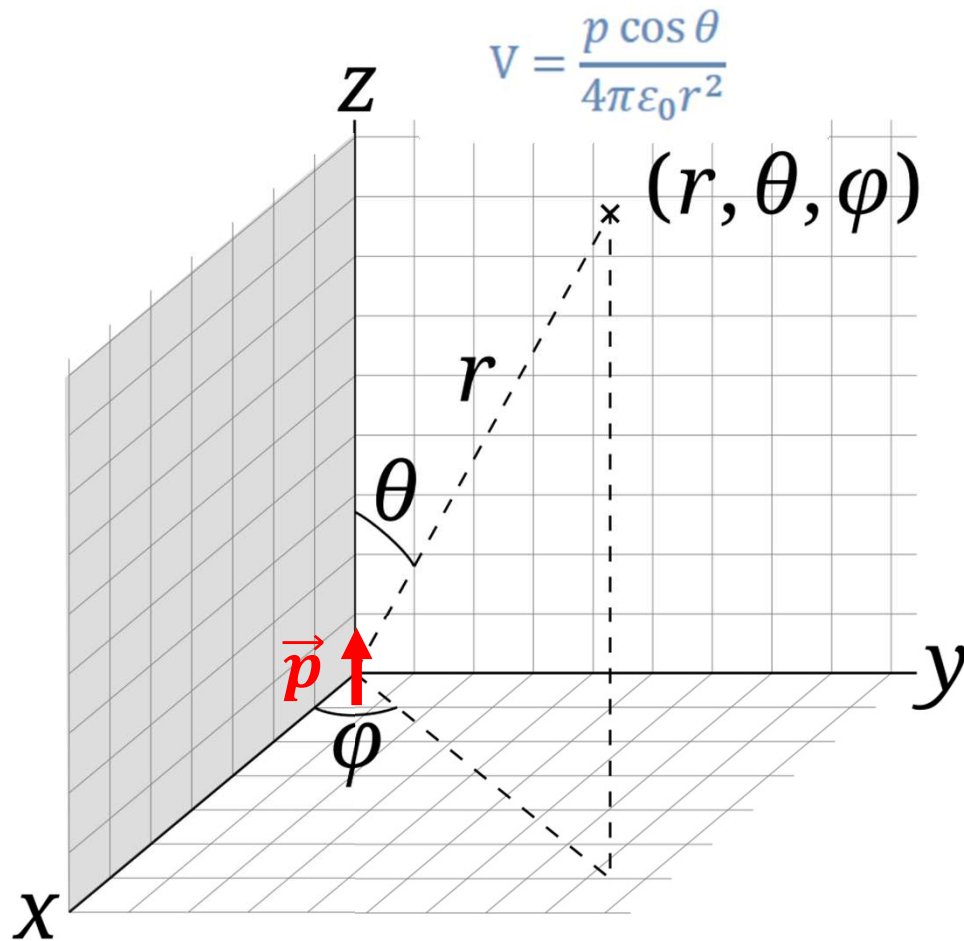


The electric field of an electric dipole



$$E_r = -\frac{\partial V}{\partial r} = \frac{p \cos \theta}{2\pi\epsilon_0 r^3}$$

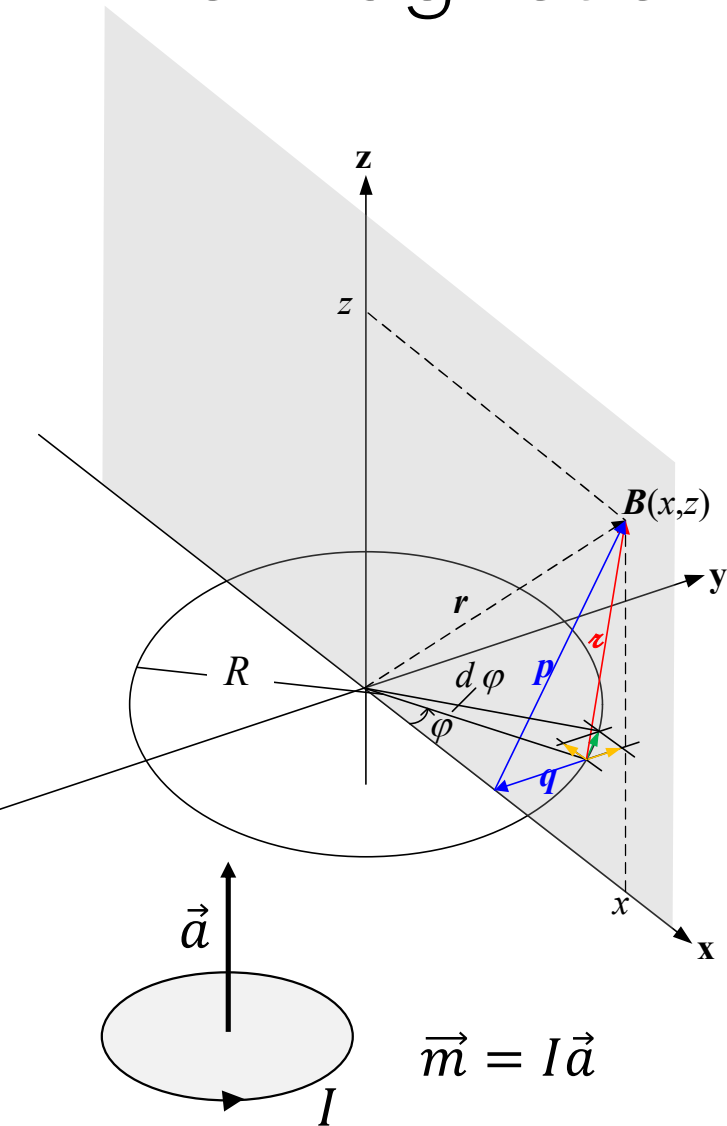
$$E_\theta = -\frac{1}{r} \frac{\partial V}{\partial \theta} = \frac{p \sin \theta}{4\pi\epsilon_0 r^3}$$

$$E_\phi = -\frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} = 0$$

On the x - z plane

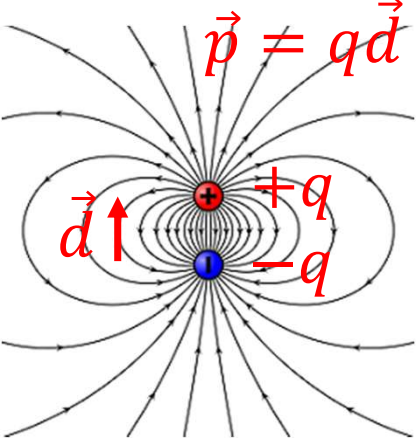
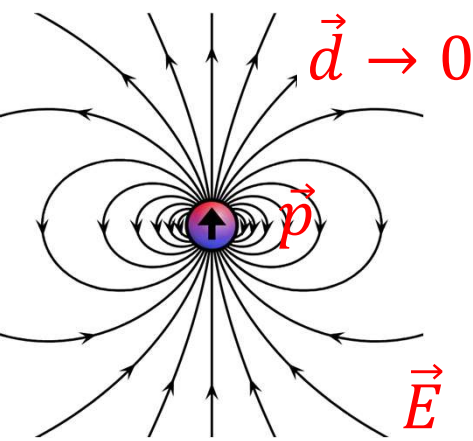
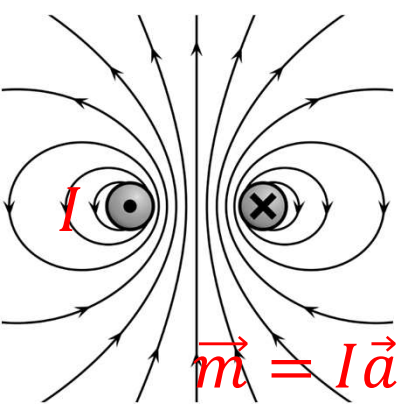
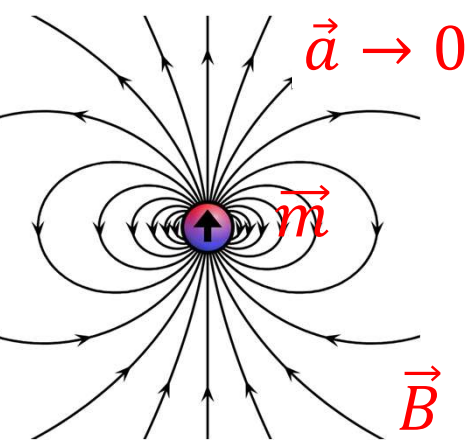
$$\begin{aligned} E_x &= E_r \sin \theta + E_\theta \cos \theta \\ &= \frac{p}{4\pi\epsilon_0} \frac{3 \sin \theta \cos \theta}{r^3} \end{aligned}$$

The magnetic field of a current loop



$$\begin{aligned}
 B_x &= \frac{\mu_0 I}{4\pi} \int_0^{2\pi} \frac{Rz \cos \varphi d\varphi}{(z^2 + (x - R \cos \varphi)^2 + R^2 \sin^2 \varphi)^{\frac{3}{2}}} \\
 &= \frac{\mu_0 I}{4\pi} \int_0^{2\pi} \frac{Rz \cos \varphi d\varphi}{(z^2 + x^2 - 2Rx \cos \varphi + \cancel{R^2 \cos^2 \varphi} + \cancel{R^2 \sin^2 \varphi})^{\frac{3}{2}}} \\
 &\approx \frac{\mu_0 I}{4\pi} \frac{Rz}{r^3} \int_0^{2\pi} \boxed{\frac{1}{\left(1 - \frac{2Rx \cos \varphi}{r^2}\right)^{\frac{3}{2}}}} \cos \varphi d\varphi \rightarrow \boxed{\approx 1 + \frac{3}{2} \frac{2Rx \cos \varphi}{r^2}} \\
 &\approx \frac{\mu_0 I}{4\pi} \frac{3R^2 xz}{r^5} \boxed{\int_0^{2\pi} \cos^2 \varphi d\varphi} \rightarrow \pi \\
 &= \frac{\mu_0 I}{4\pi} \frac{3\pi R^2}{r^3} \frac{x}{r} \frac{z}{r} = \frac{\mu_0 \textcolor{red}{I} a}{4\pi} \frac{3 \sin \theta \cos \theta}{r^3}
 \end{aligned}$$

Summary

Dipole and field patterns	Ideal dipole	E_x or B_x on x-z plane	Potentials
 $\vec{p} = q\vec{d}$	 $\vec{d} \rightarrow 0$ $\vec{p}$ $\vec{E}$	$\frac{1}{4\pi\epsilon_0} p \frac{3 \sin \theta \cos \theta}{r^3}$	$\phi = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}$ $(\vec{E} = -\nabla\phi)$
 $\vec{m} = I\vec{a}$	 $\vec{a} \rightarrow 0$ $\vec{m}$ $\vec{B}$	$\frac{\mu_0}{4\pi} m \frac{3 \sin \theta \cos \theta}{r^3}$	$\vec{A} = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}}{r^2}$ See book 5.4.3 and my class notes for proofs $(\vec{B} = \nabla \times \vec{A})$